Srikant Kharat

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PROFESSIONAL SUMMARY

Detail-oriented Automotive Diagnostics Engineer with 4+ years of experience in developing Windows-based diagnostic middleware and ECU reprogramming applications.Hands-on with UDS, DoIP and CAN/CAN-FD across the ISO 22900 / 22901 / 13209 diagnostic standards. Skilled in debugging concurrency issues and integrating diagnostic hardware interfaces to deliver high-quality automotive software.

TECHNICAL SKILLS

- **Languages:** C++, C# | C++11/14/17, Python
- **C++ & Windows:** STL, OOP, ATL-COM, MFC, Win32 API, Multithreading, Thread Synchronization, Thread Pool, Exception Handling
- **Architecture:** Layered Architecture, Factory Method, Singleton, Configuration-Driven Architecture
- **Automotive:** UDS (ISO 14229), CAN, DoIP (ISO 13400), D-PDU API (ISO 22900-2), MCD-3, SAE J2534, ODX,OTX
- **Tools:** Visual Studio, TeraTerm, CMake, Git, Vector CANoe, GTest, Wireshark

PROFESSIONAL EXPERIENCE

Onsite Technical Engineer - Tata Consultancy Services | Mar 2025 - Present | Kanagawa, Japan

Multi-Vehicle ECU Reprogramming Tool — Technical Owner **-March 2025 - Present**

- Designed a three-tier Win32 worker-thread architecture (API, ECU service and event dispatch), supporting more than ~**80 concurrent ECU workers** over a single vehicle bus.
- Implemented event-based asynchronous-to-synchronous coordination between COM callbacks and sequential UI workflows, including completion tracking, cancellation and communication-fault handling.
- Developed reference-counted worker lifecycle management to address Win32 64-handle wait limitation; implemented bounded concurrency with counting semaphores for CAN/DoIP-specific execution limits.
- Resolved production concurrency defects including deadlocks, race conditions, lost wakeups and waits

EV Battery Regulation & Data Collection Tool — Technical Architect **-April 2024 - Jan 2025**

- Designed an onboard, configuration-driven EV data collection system aligned with European regulatory requirements.
- Architected a hardware-independent diagnostic interface supporting SAE J2534-compatible devices across European markets.
- Developed a plug-and-play vehicle data acquisition and cloud-upload workflow, enabling automatic data collection without manual user interaction under the R2R regulatory framework.

Consult-III-Plus — Module Lead **-Mar 2022 - April 2024**

- Maintained 1,200-file ATL-COM middleware stack integrating WPF, COM drivers, MCD-3 diagnostic engine, and third-party D-PDU APIs, ensuring seamless interoperability.
- Implemented extensible diagnostic components using Strategy, Command, Template Method, Adapter and Facade patterns.

ECU Reprogramming Tool — C++ / C# Developer **-Mar 2022 - Feb 2023**

- Decoupled UI and middleware using asynchronous processing and thread-pool execution, improving tool performance by **40%**.
- Designed a data-independent diagnostic communication approach, eliminating the need for predefined ECU-specific comm data.

ECU Simulation & Validation Toolchain

- Configured and validated **50+ simulated ECUs** and benchmarked gateways using Vector CANoe, vTESTstudio and Vector hardware.
- Performed UDS-over-CAN/DoIP diagnostic validation and cross-ECU fault analysis

EDUCATION & CERTIFICATIONS

B.E. Computer Science & Engineering -

MGM's College of Engineering, Nanded, India | CGPA 9.1 /10 -2016 - 2021

Crash Course on Python | 2022

ACHIEVEMENTS

- **On-the-Spot (Team) Award**, Tata Consultancy Services - for delivery of the Multi-Vehicle Reprogramming Tool (5,000+ vehicles reprogrammed in one year, under 2% error rate).
- Selected for **onsite deployment to Japan** as the technical interface between the customer's engineering team and offshore development.